Framework and Methods of State Monitoring-Based Positioning System on WIFI-RTT Clock Drift Theory

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High-precision indoor positioning problems have attracted considerable attention recently. The indoor ranging and positioning method based on wireless fidelity (WIFI) round-trip-time has become popular. However, resulting from clock drift (CD), there are two problems that are still not completely resolved and need to be addressed. The first problem is ranging error drift, namely the reset of phase distortion errors (PDEs), while the second problem is the communication state change, namely ranging information loss and restoration. To solve these two problems, this study noticed that the reset of PDEs will only change along with the changes in communication states. Thus, a monitoring-based positioning system (MPS) framework is presented based on the theoretical model of CD as a method to account for ranging error drift and communication state change when estimating position with the IEEE 802.11mc protocol. Then, a state monitoring algorithm (SMA) suitable for the MPS framework is proposed to accurately capture the moment of communication state change. Further, a PDE constraint (PDEC) model is proposed based on the SMA to estimate PDEs in different communication states to reduce the impact of CD on positioning results. The experimental results finally show that considerable improvement in positioning accuracy can be obtained using the SMA and PDEC model compared with that obtained using the semiparametric solution algorithm.

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I. INTRODUCTION

Along with the increase in location-based services, wireless fidelity (WIFI), Bluetooth, ultrawideband, inertial navigation, and other signal sources are helpful for indoor positioning [1], [2], [3], [4], [5], [6], [7], [8], [9]. Moreover, WIFI round-trip-time (WIFI-RTT) has attracted more attention for public usage [10], [11], [12], [13], [14]. The superiority of this method has attracted research attention to indoor positioning using WIFI as a wireless signal source [15], [16], [17].

Recently, researchers have explored and applied the WIFI-RTT ranging method [18], [19]. However, two problems of ranging error drift and communication state change still need to be solved completely and have received little attention in the literature. The overall numerical significance will occur in the positioning results when ignoring the impact of clock drift (CD) on RTT ranging and positioning. This is to say, the bias degree changes when the communication state changes. Changes in the bias center accompany communication state changes. Fig. 1 illustrates the degree of deviation between the static positioning results ignoring the impact of CD and the standard position results.

Previous research believes that the ranging error drift is one of the results of the multipath effect due to the WIFI signal's transmission characteristics [11], [20]. Guo et al. [21] have developed a method using a Kalman filter to fuse calibrated RTT ranging and RSS data. Cao et al. [22] used Cubic fits to correct errors. But they overlooked the impact of the CD on the ranging information, so the effect of correction is very limited.

The WIFI-RTT positioning method can be designed by a semiparametric model [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35]. The semiparametric model does not need to consider the change time of communication state and phase distortion error (PDE) characteristics and needs to solve the PDE in each epoch. This is a mathematical method based on statistical theory, meaning the errors it corrects should follow a Gaussian distribution. But the PDE is a step fault, which is nonlinear and non-Gaussian [36]. Thus, this method needs to improve the accuracy of positioning results.

Using the theoretical model of clock drift (CD model) as the theoretical basis, acknowledging the physical and statistical characteristics of PDE, this study aims to propose a state monitoring algorithm (SMA) and PDE constraint (PDEC) model suitable for the monitoring-based positioning system (MPS) framework to reduce the influence of PDE on positioning results. The significant contributions of this study are summarized as follows. First, this study proved that the data upload period does not affect ranging accuracy. This study provides evidence that when the data upload period is changed, the positioning accuracy is unaffected. Second, an MPS framework based on the CD model is proposed. This study provides a reference template and an

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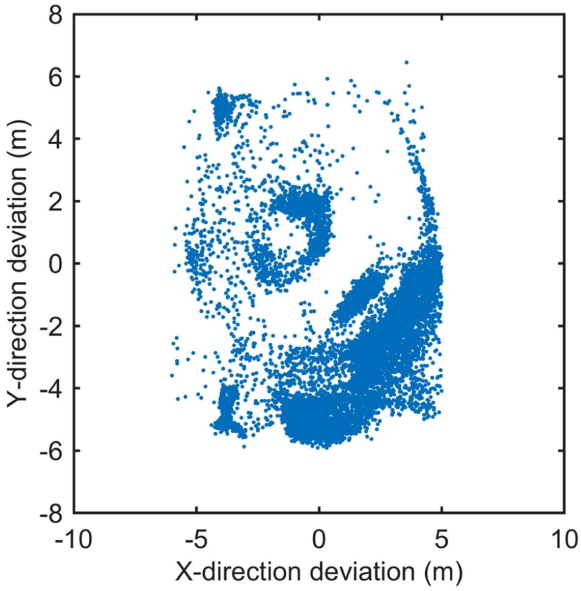

Fig. 1. Degree of deviation between the static positioning results ignoring the impact of clock drift and the standard position results.

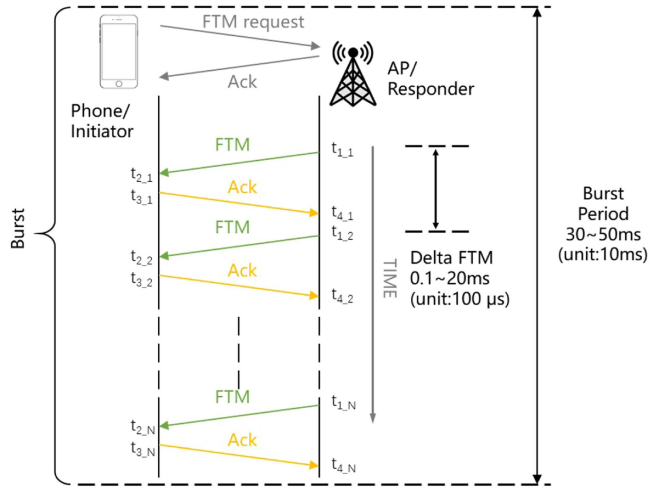

Fig. 2. WIFI-RTT ranging principle and the needed time for communication in each stage.

example for future positioning systems based on the CD model. Third, an SMA suitable for the MPS is proposed to capture the moment when the communication status changes accurately. This study successfully transforms the CD model into reality and provides a basis for correcting the PDE at the initial stage of state change. Finally, a PDEC model based on the SMA is proposed to correct the PDE.

In this study, an estimation model that relies on the statistical characteristics of PDE and the change rule of communication state is first established, which greatly improves the positioning accuracy. MPS, SMA, and PDEC are the core innovative contributions of this article. This study performed experiments to validate the aforementioned proposals. In future work, the research group would focus on the influence of the communication process on ranging results and try to propose a method to determine the PDEs faster than the PDEC model.

II. METHODOLOGY

In this section, the RTT ranging principle and the impact of the data upload period on running results will be described first. Then, the difficulty of reducing the influence of PDE on positioning results is analyzed. After that, the MPS framework is described in detail. Then, the SMA model, PDEC model, and the least-squares localization algorithm suitable for the MPS framework are introduced. MPS, SMA, and PDEC are the core innovative contributions of this article.

A. Impact of the Data Upload Period on Ranging Results

As concerned by the CD model, the data upload period determination affects the communication process, whereas the influence of the data upload period on ranging results has not been determined yet. Therefore, this problem is addressed in this subsection. The IEEE 802.11mc protocol confirms the principle of WIFI-RTT ranging (see Fig. 2). Based on data and experimental verification, the ranging formula is expressed as follows [37], [38], [39]:

$$\begin{cases} d_{\mathrm{RTT}} = \frac{1}{2} \cdot \mathrm{RTT} \cdot c + d_{\mathrm{com}} - d_{\mathrm{trans}} - d_{\mathrm{MPE}} \\ \quad -d_{\mathrm{NLoS}} + a + \varepsilon \\ \mathrm{RTT} = \frac{1}{N}\left(\sum_{i=1}^{N} t_{4_i} - \sum_{i=1}^{N} t_{1_i}\right) \\ \quad -\frac{1}{N}\left(\sum_{i=1}^{N} t_{3_i} - \sum_{i=1}^{N} t_{2_i}\right) \end{cases} \tag{1}$$

where d_{RTT} denotes the distance between a router and phone, RTT denotes the average cycle time obtained by phone; d_{com}, d_{trans}, d_{MPE}, and d_{NLoS} represent errors caused by the communication protocol, signal propagation in the hardware, multipath effect, and non-line-of-sight phenomena, respectively; a represents the PDE at the current communication state; ε represents the speed of light in vacuum; t_{1_i}, t_{2_i}, t_{3_i}, and t_{4_i} represent the timestamp provided by the router (AP) and phone at different stages, and their specific meanings are shown in Fig. 2; N represents the number of successful WIFI-RTT ranging; and the default value is 8 limited by the development environment.

WIFI-RTT is a ranging method relying on communication. Its ranging accuracy can be kept unchanged on the premise of ensuring the integrity of the communication process. According to the WIFI-RTT ranging principle, a ping pong protocol communication time (a delta fine-time measurement (FTM)) is greater than 20 ms. Multiple delta FTMs constitute the whole burst period. Therefore, only if the data upload period is set to exceed the entire burst period the ranging accuracy will remain the same [11].

B. Difficulties

We investigated the impact of CD on WIFI-RTT positioning. And then, we stated the root cause of the ranging error drift and communication state change by proposing a theoretical model of CD [36].

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PDE is a semisystematic error that changes with the change of communication state. PDEs vary with the communication state, showing stability in the same communication state and randomness in different communication states. It is determined by both the frequency and phase of the transmitter's oscillator. The phase difference leads to the appearance of the PDE, and the phase reset leads to the variation of the PDE. This is also the cause of the ranging error drift.

While communication state change, which is ranging from information loss and recovery, is manifested during mobile devices processing cycles of communication packets from different routers to overlap or separate caused by the CD. The CD changes the communication status, resulting in a change in the phase of the oscillator, which is the cause of the ranging error drift.

According to the theoretical model of CD, the PDE varies due to the difference between receivers, routers that can successfully communicate with receivers, and communication states. In other words, the PDEs of different users are varied with no correlation. The error solutions from one user are not able to correct the ranging data of another user, which means that the PDE of a certain user at a fixed time is unique. Therefore, the accuracy and stability of the positioning results can only be ensured by the application of the first epoch data when solving the PDE after the communication state changes. It is a necessary but difficult problem to reduce the influence of CD on WIFI-RTT positioning results, which quickly capture the moment of communication state change and correct the PDE in a short time.

C. MPS Framework

To avoid the positioning accuracy problem caused by PDE of inadequate absorption as well as the waste of computing resources caused by PDEs, which must be solved in each epoch, the physical characteristics of PDE and the design of a set of processing methods for solving PDE need to be considered from the beginning. Obviously, analyzing PDE as an independent error source and solving it separately is very helpful in understanding the PDE space. Compared with the rough processing of other errors simultaneously, the positioning accuracy will be greatly improved. According to the theoretical model of CD, PDE is a step fault, which changes immediately when the communication state changes until it enters the next communication state. Therefore, according to the physical characteristics of PDE, the error can be fixed at the early stage of error reset. This processing strategy is more complex than the semiparametric model; however, considering that it reduces the solving frequency of PDE from every epoch to every communication state change, the impact of computing time on localization efficiency needs to be addressed.

The following scheme is formulated in this study based on the aforementioned ideas. First, capture the time when the communication state changes. Then, the PDE values are calculated. After, the data of PDE in the communication state are compensated to obtain the final positioning results. The proposed method's framework is shown in Fig. 3. The characteristics of this framework compared with other positioning systems are as follows. First, SMA is proposed through the CD model, SMA can feedback on a change of communication state through data information and send alarm information to the positioning model and the PDEC model in time. In other words, the inputs of SMA are the RTT-ranging results of this epoch. If the communication state is judged to change, the alarm signal will be output; otherwise, there is no output. Therefore, SMA is active throughout the process. The PDEC model relies on the statistical characteristics of PDEs, and it only starts working after receiving the alarm information provided by SMA. The inputs are RTT ranging results in this epoch, while the outputs of the PDEC model are the PDE values from each router. The workflows of the SMA and PDEC models are described in detail in the following two sections. Finally, the location is solved by using the ranging information after the PDE is corrected, and then, the positioning result is output and evaluated.

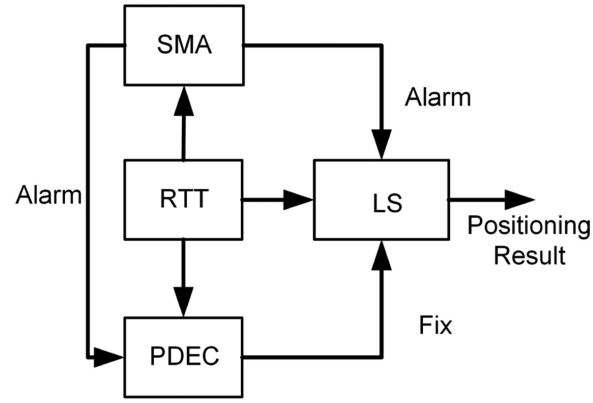

Fig. 3. MPS framework based on the CD model.

1) *SMA Suitable for the MPS Framework:* The PDE changes with the communication state. In other words, the moment when the communication state changes is when the PDE changes. Therefore, it succeeded in turning the problem of searching for the reset time of PDE through data processing into the problem of obtaining the change time of communication state through monitoring. By monitoring the communication state, the change time of the communication state can be accurately obtained, and the reset time of the PDE can be obtained simultaneously. An SMA suitable for the MPS framework is proposed to solve the aforementioned problem, and Algorithm 1 shows the pseudocode of SMA.

The best indicator of changes in a communication state is whether a new router communicates with the receiver successfully. If a new router appears, the communication state must change. If no new router appears, further identification and verification are required. From Algorithm 1, the cumulative time threshold $lineaddtime$ and the time difference threshold $linet^d$ need to be initialized according

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Algorithm 1: SMA Suitable for the MPS Framework.

0 Initialize: $flag = 0$, $alarm = 0$, $linet^d$, $lineaddtime$, FTT
for i
1 Get RTT data: t_i, N_i
for $j = 1 : N_i$ **do**
  Mac_i^j, RTT_i^j, PDE_{i-1}^j, Poi_i^j **end for**
get other data: $addtime_{i-1}$
2 Time calculation:
$t_i^d = t_i - t_{i-1}$, $addtime_i = addtime_{i-1} + t_i^d$
if $t_i^d > linet_i^d$ **do**
$flag = 1$; **end if**
if $addtime_i > lineaddtime$ **do**
$alarm = 1$; $addtime_i = 0$; **end if**
3 Discrimination**:**
if $flag == 1 \&\& alarm == 0$ **do**
if $N_i > N_{i-1}$ **do**
    $alarm = 1$; $addtime_i = 0$;
  else
    if $\{\mathrm{Mac}_i^j\} \not\subset \{\mathrm{Mac}_{i-1}^j\}$ **do**
      $alarm = 1$; $addtime_i = 0$;
    else
$C = f(\{\mathrm{PDE}_{i-1}^j\}, \{\mathrm{RTT}_i^j\}, \{\mathrm{Poi}_i^j\})$
      if $C > \mathrm{FFT}$ **do**
        $alarm = 1$; $addtime_i = 0$;
      end if
      end if
  end if
if $flag == 0 \&\& alarm == 0$ **do**
    if $N_i > N_{i-1}$ **do**
      $alarm = 1$; $addtime_i = 0$;
  else
      if $\{\mathrm{Mac}_i^j\} \not\subset \{\mathrm{Mac}_{i-1}^j\}$ **do**
$alarm = 1$; $addtime_i = 0$;
endif
end if
end if
4 Out: $alarm$
0 Initialize: $flag = 0$, $alarm = 0$
end for

to the data upload period, and then, the maximum fault-tolerant cost, i.e., fault-tolerant threshold (FTT) should be determined. Here, superscript j represents different routers and subscript i represents the current moment (the same as follows).

$lineaddtime$ describes the maximum detection period to ensure that periodic PDEs meet the positioning accuracy requirements. The reason for this is that when the data upload period is large, the accumulated noise is the main factor in changing the communication state, so the time difference cannot be used as the only indicator for further inspection. One parameter associated with $lineaddtime$ is $addtime_i$, which adds up each actual time difference to represent the cumulative time of time I and reset to 0 each time an alarm is triggered. Whenever $addtime_i$ exceeds $lineaddtime$, an alarm is triggered, i.e., alarm = 1. In other words, this event has the highest priority.

$linet^d$ describes the maximum acceptable CD. It is related not only to the set data upload period but also to the mean and variance of the time difference of the current communication state, which is an adaptive result. If the current time difference t_i^d exceeds $linet^d$, it is considered that there is a greater possibility of communication state change; thus, further detection is required, namely, $flag = 1$.

FTT describes the tolerance of the maximum acceptable cost function. Assuming that the communication state does not change, the PDE of the previous moment $\{\mathrm{PDE}_{i-1}^j\}$ is corrected to the sampling data $\{\mathrm{RTT}_i^j\}$ corresponding to the current router Mac address $\{\mathrm{Mac}_i^j\}$. Then, the router position information $\{\mathrm{Poi}_i^j\}$ was used to construct the data matrix B, residual vector L, and the parameter vector to be estimated x, and the form is shown as (4) (the same below). The current cost function value C was calculated by the least square cost function. If it is bigger than FTT, the communication state is considered to have changed.

In addition, with the SMA, dynamic positioning could then be separated from static positioning. The advantage is that the FTT of the two cases can be determined separately, and the FTT needs to be adjusted as follows. In the static positioning process, FTT is considered to be a fixed value; meanwhile, in the dynamic positioning process, the last five-position data are called simultaneously to obtain the movement information of the target. FTT is proportional to the motion information of the object.

2) *PDEC Model Suitable for the MPS Framework:* This section presents a PDEC model applicable to the above framework. After obtaining the SMA alarm signal, the PDEC model should be applied to the new communication state. The PDE is expected to be estimated quickly (much less than 1 s) after the communication state changes. Superscripts m and n both represent the number of solutions, superscript j represents different routers, and subscript i represent the current number. The design idea is as follows.

It is confident that the more accurate the PDE is for the same data, the smaller the cost function value. Because the statistical characteristics of PDE are mastered, the PDEs can be corrected directly according to the ranging results. Set the value of repair as a, i.e., take $\mathrm{RTT}^j + \mathrm{PDE}_{i-1}^j$, $\mathrm{RTT}^j + \mathrm{PDE}_{i-1}^j + a_i$, and $\mathrm{RTT}^j + \mathrm{PDE}_{i-1}^j - a_i$ as the ranging result. In addition, the position result $\{\mathrm{all}_i^n\}$ and corresponding cost function value $\{\mathrm{cost}_i^n\}$ are calculated.

An enormous amount of data need to be corrected, i.e., there will be an immense cost function value; this cost function value of the extensive data is obviously unsuitable. Then, the calculated position results are sorted according to the cost function value, and the available position result $\{\mathrm{cor}_i^m\}$ and its corresponding cost function value $\{\mathrm{af\cos t}_i^m\}$ are obtained. Let the number of available positional solutions be b_i of the number of positional solutions $\{\mathrm{all}_i^n\}$, i.e., $m = n * b_i$, where $n = 3^x$, x indicates the number of routers, and b_i indicates the proportion of functional solutions to total solutions. In this study, the cost functional results are

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Algorithm 2: PDEC Model Suitable for the MPS Framework.

0 Initialize: $c, d, W, \{\mathrm{PDE}_0^j\} = \{0\}$
for $i = 1 : W$ **do**
1 Get RTT data: t_i, N_i
for $j = 1 : N_i$ **do**
$\mathrm{Mac}_i^j, \mathrm{RTT}_i^j, \mathrm{PDE}_{i-1}^j, \mathrm{Poi}_i^j$ **end for**
get other data: a_i, b_i
2 Data calculation: $\mathrm{RTT}^j + \mathrm{PDE}_{i-1}^j$,
$\mathrm{RTT}^j + \mathrm{PDE}_{i-1}^j + a_i, \mathrm{RTT}^j + \mathrm{PDE}_{i-1}^j - a_i$
3 Positioning calculation:

$\{\mathrm{all}_i^n, \cos \mathrm{t}_i^n\} = \mathrm{LS}(\{\mathrm{RTT}_i^j, \mathrm{RTT}^j + \mathrm{PDE}_{i-1}^j + a_i,$
$\mathrm{RTT}^j + \mathrm{PDE}_{i-1}^j - a_i\}, \{\mathrm{Poi}_i^j\})$
4 Sort and select:
$\{afall_i^n, \mathrm{af}\cos \mathrm{t}_i^n\} = \mathrm{sort}(\{\mathrm{all}_i^n, \cos \mathrm{t}_i^n\});$
$\{\mathrm{cor}_i^m, \mathrm{af}\cos \mathrm{t}_i^m\} = s(\{afall_i^n, \mathrm{af}\cos \mathrm{t}_i^n\}, b_i)$
5 Reference position calculation:
$\{w_i^m\} = \mathrm{normalize}(\{\mathrm{af}\cos \mathrm{t}_i^m\});$
$P_i = wm(\{w_i^m\}, \{\mathrm{cor}_i^m\})$
6 PDE calculation:
$a_{i+1} = c * a_i$
$b_{i+1} = d * b_i$
$\{\mathrm{PDE}_i^j\} = g(P_i, \{\mathrm{PDE}_{i-1}^j\}, \{\mathrm{RTT}_i^j\}, \{\mathrm{Poi}_i^j\})$
end
7 Out: $\{\mathrm{PDE}_w^j\}$

taken as the weighting basis, and the weight is considered to be negatively correlated to the cost function value corresponding to the normalized available position solution. Then, the public position solution is the weighted average, and the weighted average result is taken as the reference position. After the reference position of the client is obtained, the PDEs between the client and individual routers can be easily obtained. Parameters a and b are modified, set $a_{i+1} = c * a_i$, $b_{i+1} = d * b_i$, and resubstitute a_{i+1} and b_{i+1} into the aforementioned steps, repeat the loop many times, and consider the final output to be ideal. Algorithm 2 shows the pseudocode for the PDEC model.

Notably, c is an attenuation factor: as the number of iterations increases, the degree of correction decreases; and d is an enhancement factor: as the number of iterations increases, the number of trusted data increases.

3) *Localization Algorithm Used in MPS:* After obtaining a router location and the distance between a phone and a router, the measurement equation can be established. Through Taylor series expansion, the indirect adjustment equation can be obtained as follows,

$$L = Bx \tag{2}$$

the Taylor series expansion is used to obtain the aforementioned indirect adjustment of the equation. Real-time positioning can be achieved after receiving the information of more than three routers as,

$$x = (B^T B)^{-1} B^T L \tag{3}$$

where two dimensions were considered. Assuming that there are m routers, the parameter expression is constructed by RTT ranging information and PDE as follows:

$$\begin{cases} L = \begin{bmatrix} d_{\mathrm{RTT}}^1 - d_1 + a \\ \vdots \\ d_{\mathrm{RTT}}^m - d_m + a \end{bmatrix} \\ \hat{x} = \begin{bmatrix} x & y \end{bmatrix}^T \\ B = \begin{bmatrix} \frac{x_1 - x_0}{d_1} & \frac{y_1 - y_0}{d_1} \\ \vdots & \vdots \\ \frac{x_m - x_0}{d_1} & \frac{y_m - y_0}{d_1} \end{bmatrix} \\ d_i = \sqrt{(x_i - x_0)^2 + (y_i - y_0)^2} \end{cases} \tag{4}$$

where a denotes the system error, the base station position is $\mathrm{AP}_i(x_i, y_i)$, the approximate position of the receiver to be obtained is $P(x_0, y_0)$; $\hat{x}$ denotes the positioning result correction; and B denotes the data matrix constructed based on location information. In the natural calculation process, the *position* correction result can be obtained using (4).

III. EXPERIMENTS AND RESULTS

This section first describes the experimental setting. Second, the impact of the data upload period on the ranging results is clarified through experimental results. Subsequently, monitoring experiments are performed to determine the correctness and practicability of the SMA. As it is challenging to verify the success rate of the PDEC model quickly, the positioning results are obtained in the last part, which applies both the SMA and PDEC model (MPS algorithm) to real-time positioning experiments simultaneously. Then, the results are selected as the basis for verifying the success rate of the PDEC model. The positioning results of the semiparametric method are considered reference results. To evaluate the practicality of the MPS algorithm, we conduct a comparative analysis of static positioning results achieved in both ideal and natural environments, as well as dynamic positioning outcomes within the natural environment. Our assessment employs the mean deviation (ME), standard deviation (STD), root mean square (RMS), and operation time as the performance metrics.

A. Experimental Setting and Accuracy Calibration

The demonstration of the experiment is shown in Fig. 4. The ideal environment is the *open* environment. An open field was set to mitigate the impact of the multipath effect on the ranging results for highlighting the impact of CD. Using pixel 4 produced by Google as initiator and WILD routers made by Compulab as responders, one mobile phone device and four routers were all placed on tripods standing at the same height (1.4 m). The routers are set in the corners of a rectangular, while the mobile phone is in the center area of this rectangular.

The routers' reference locations were their external antenna centers, and the phone's reference location was their geometric center. In the process of ranging, only one router

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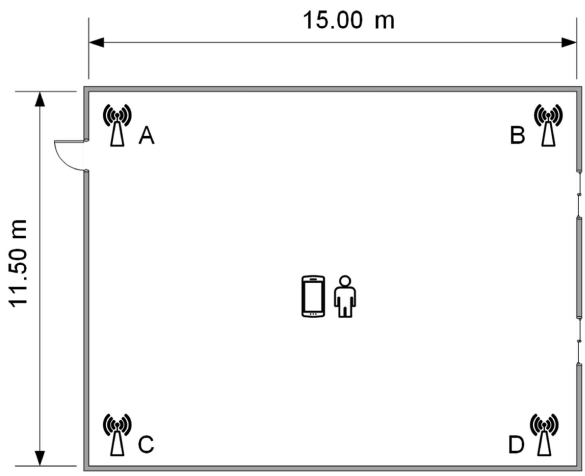

Fig. 4. Experimental setting with four routers and one mobile phone in a rectangular layout.

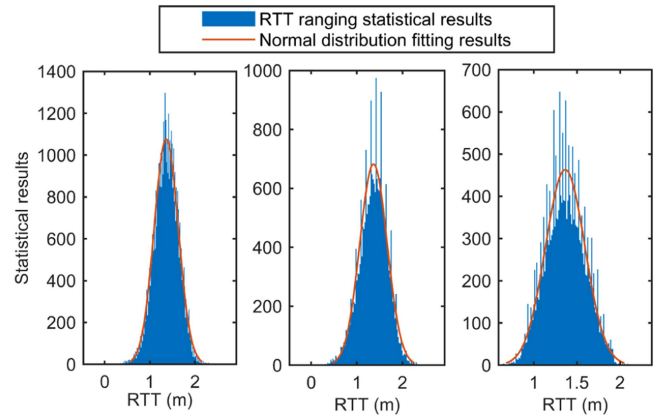

Fig. 5. Data statistical results and normal distribution of fitted curves under different data upload periods. From the left panel to the right panel are 200, 600, and 1000 ms.

was turned ON to prevent the impact of CD. The change of communication state is *only* set to be triggered by the mobile phone restart process. However, in the positioning process, all routers were turned ON to highlight the impact of CD on ranging and positioning results. A constant system error is also set in the position calculation considering the irretrievable hardware damage to ensure data calibration accuracy [9].

B. Experimental Verification of Data Upload Period on Ranging Results

As mentioned previously, this study described the WIFI-RTT ranging principle and explained the relationship between the data upload period and ranging accuracy. This study determines the impact of different data upload periods on running accuracy in this subsection by analyzing the data collected under three sets of experiments. The experimental scenario is shown in Fig. 4. A router and phone were placed on a tripod of equal height, with a distance of 6.4 m between the two. Changes were detected in the ranging results under different data upload periods. To reduce the influence of PDE and random errors in the ranging results, each experiment contained 20 groups with a ranging period of 10 min. Finally, all data obtained under the same period are processed simultaneously. The statistical results are shown in Fig. 5. The data upload periods of the three experiments from left to right are 200, 600, and 1000 ms. It can be concluded from the figure that although the data upload periods differ, there is no noticeable difference in the mean and variance of the results obtained in each group. The indicators are approximately equal when the amount of data is sufficient. From this, adjusting the data upload period does not affect the ranging accuracy when the data upload period exceeds the necessary range time.

C. Evaluation of the SMA

The operating mechanism of the SMA is described before. In this subsection, the SMA is evaluated through experimental data. The specific evaluation mechanism is divided into the following parts.

First, this study evaluates the static and dynamic experiments separately because the vigorous investigation has more complicated environmental state changes than the static experiment. The CD model illustrates how the complex environment changes the communication state faster. Second, the rate of successful and false alarms need to be evaluated. When the communication state changes and the alarm is triggered successfully, it is called a successful alarm; the ratio of the number of successful alarms to the number of communication changes is called the success alarm rate. Meanwhile, if the communication state changes but the alarm is not triggered, it is called a false alarm; the ratio of the number of false alarms to the number of communication changes is called the false alarm rate. Third, it is necessary to evaluate the reasons for the changes in the communication state caused by the time difference sudden change (mode 1) and the cumulative time difference exceeding the FTT (mode 2) because it can be seen from the CD model that more attention was paid to the phenomenon of sudden time differences.

Meanwhile, there was no suitable technique to handle the change in communication state caused by mode 2, so this study can only rely on periodic verification to remedy this. From the aforementioned analysis, the SMA has better state monitoring effects on small data upload period data than on extensive data upload period data. Therefore, at the end of the experiment, this research suggested some data upload period impacts.

In the experiment in this section, 150 communication state change times were collected for each state as a judgment sample. The average of the three sample sets was used as the final result. Table I describes the SMA's success and false alarm rates in modes 1 and 2 in the static and dynamic positioning results. The data upload period used in the experiment is 500 ms. Table II describes the SMA's success and false alarm rates at different data upload periods. The data upload periods used in the experiments are 200, 500, and 1000 ms. From the data in Tables I and II, the SMA

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TABLE I
Usability Evaluation of SMA in Different Modes

Mode \ Experimental method		Static positioning	Dynamic positioning
Mode 1	success alarm rate	99.7%	98.6%
	false alarm rate	0.3%	1.4%
Mode 2	success alarm rate	86.3%	92.4%
	false alarm rate	13.7%	7.6%
Total	success alarm rate	98.3%	98.1%
	false alarm rate	1.7%	1.9%

TABLE II
Usability Evaluation of SMA in Different Data Upload Periods

Data upload period \ Experimental method		Static positioning	Dynamic positioning
200 ms	success alarm rate	99.8%	99.6%
	false alarm rate	0.2%	0.4%
500 ms	success alarm rate	98.3%	98.1%
	false alarm rate	1.7%	1.9%
1000 ms	success alarm rate	96.8%	95.9%
	false alarm rate	3.2%	4.1%

has high practical value and can keenly capture the moment when the communication state changes.

From Table I, the success alarm rate of mode 1 of the dynamic experiment is lower than that of the static experiment. This situation is well-understood because the vigorous investigation complicates the communication state, and there are some small state changes. It does not reach the preset threshold in the monitoring strategy and cannot be effectively captured. However, the success alarm rate of mode 2 of the dynamic experiment is higher than that of the static investigation. This is because dynamic testing makes more state changes determined by the time difference sudden change. Compared with static positioning, the proportion of mode 1 was 78.6% and this increases to 87.2% under dynamic positioning. Therefore, a small base brings a higher ratio.

The results in Table II show that with the shortening of the data upload period, the success alarm rate of SMA increases significantly. This is because the SMA is better at listening for state changes caused by the time difference sudden change. According to the CD model, the shorter the data upload period, the higher the probability of state change caused by the time difference sudden change. Therefore, a faster data upload period corresponds to a higher detection success rate regardless of the static or dynamic states. This also explains the correctness of the theoretical model of CD. Thus, when the SMA is used, the data upload period of the data should be adjusted as high as possible, which also agrees with the sampling strategy for improving the ranging accuracy.

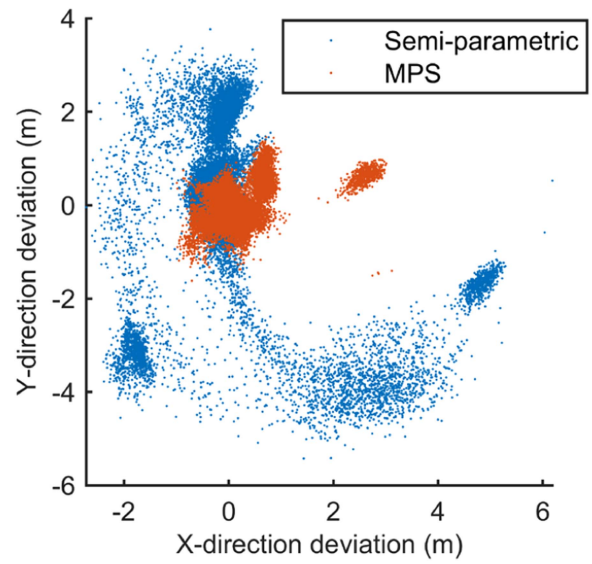

Fig. 6. Deviation value between static positioning result the under the ideal environment obtained through MPS and semiparametric method and standard position.

D. Static Positioning Effect of Indoor Positioning Algorithm Based on MPS Under Ideal Environment

This section mainly introduces the positioning effect of indoor positioning based on the MPS algorithm. To evaluate the performance of the proposed method, some experiments were performed. In this experimental part, the data upload period was set to 200 ms; the scene settings are shown before. Two positioning methods are used to estimate the phone position: the least square (LS) algorithm based on MPS and the semiparametric model. Calculating the positioning results within 1 h under the two methods. The deviation results between static positioning result obtained through the two method and standard position are shown in Fig. 6. The positioning effect of the MPS algorithm is much better than that of the semiparametric model.

The positioning effects of the semiparametric model and MPS algorithm are shown in Fig. 7. The maximum positioning errors of the MPS and semiparametric methods on the x-axis are 3.11 and 6.19 m, respectively; and the maximum errors on the y-axis are 1.6 and 5.43 m, respectively. From the x-axis, these two methods have similar effects because valid outliers have been treated as invalid outliers in the calculation process. Overall, the semiparametric model has more outliers than the MPS method, which shows that the

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TABLE III
Data Statistics of the Static Positioning Results Under the Ideal Environment Obtained Using Two Methods

Method \ Precision	ME		STD		RMSE	
	X	Y	X	Y	X	Y
Semiparametric	0.08 m	−0.17 m	1.31 m	1.78m	1.35 m	1.79 m
MPS	0.08 m	0.02 m	0.55 m	0.38 m	0.58 m	0.39 m
Promotion	0	88.24%	58.02%	78.65%	57.04%	78.21%

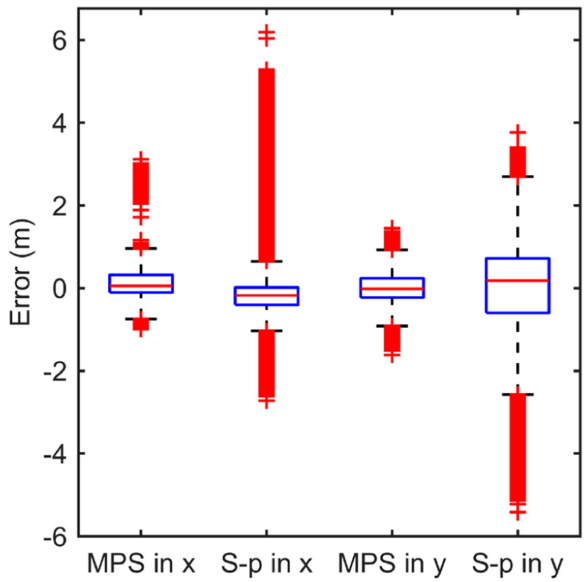

Fig. 7. Box diagram of static positioning error results under the ideal environment of MPS and semiparametric method in *X* and *Y*-axes.

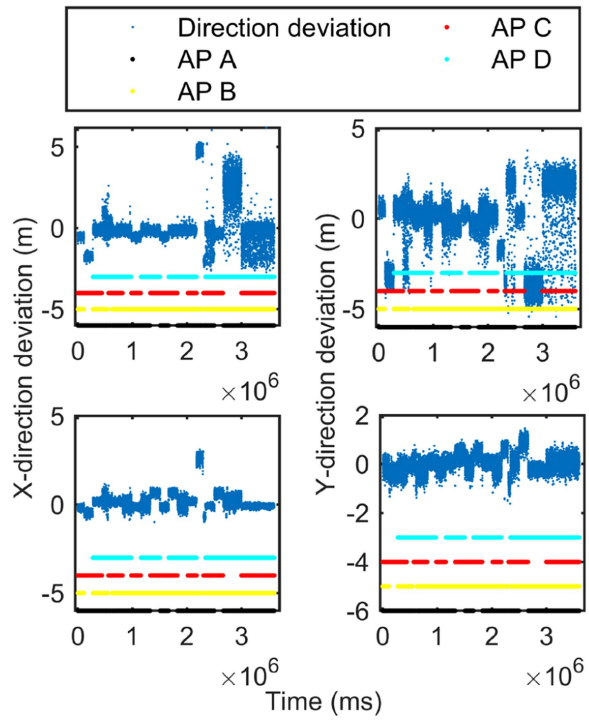

Fig. 8. Changes in the deviation of the static positioning results under the ideal environment of the two methods over time. The left panel shows the variation in the *x*-axis, while the right panel shows the divergence in the *y*-axis. And the upper panel shows the result of the semiparametric model, while the lower panel shows the effect of MPS.

proposed method is much better than the semiparametric model.

Table III shows the ME, STD, and RMS error (RMSE) based on the semiparametric model and MPS algorithm. Compared with the semiparametric model, the MPS method has advantages regarding ME, STD, and RMSE. Because RMSE can show localization stability to a certain extent, the MPS method has better localization stability than the semiparametric model. The positioning error of the MPS method is always lower than that of the semiparametric model.

In Fig. 8, the blue point on the upper two panels and the lower two panels describe the variation of the deviation degree of the positioning results of the semiparametric model and MPS method in the *x*- and *y*-directions over time. And the four discontinued parallel lines (labeled AP A, AP B, AP C, and AP D) refer to the router labels that successfully communicate with the mobile device at different times. Through the positioning results of the two models, the deviation degree in the *x* and *y*-directions changes with time. The positioning results of MPS in each communication state cycle exceed the solution results of the semiparametric model.

In addition, it can be found that in a period when the positioning error is substantial, the receiver only obtains the information of two routers. Although MPS does not fail, a result related to the solution result of the previous epoch is given, but the degree of error correction is weak.

It shows that the positioning result still has periodic jumps even after the PDE correction. By comparing the label of routers marked in Fig. 8, it shows how the distribution of routers causes these fluctuations. Because the distribution of routers is not uniform, it is equivalent to the weight distribution of data in different directions in the process of positioning solution, which causes the jump in positioning results. The problem of positioning errors caused by uneven router distribution is also the focus of our future research.

The aforementioned results show that the application based on the MPS method can obtain reliable location results effectively. To study the computational efficiency of this method, a computer with Intel 11 CPU 8-GB memory and Windows 10 operating system were selected as the test platform to obtain the running time. The average

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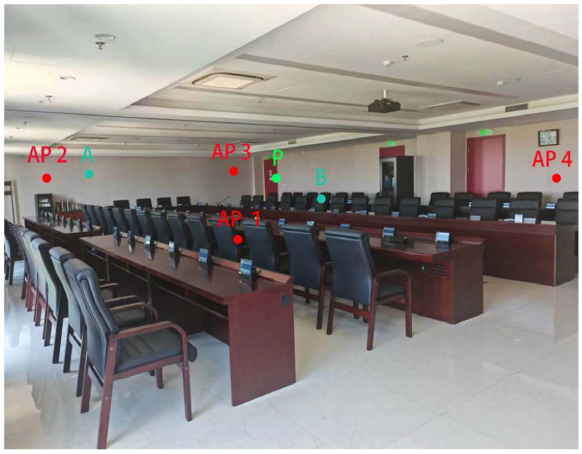

Fig. 9. Scene settings (the conference room selected as the natural environment).

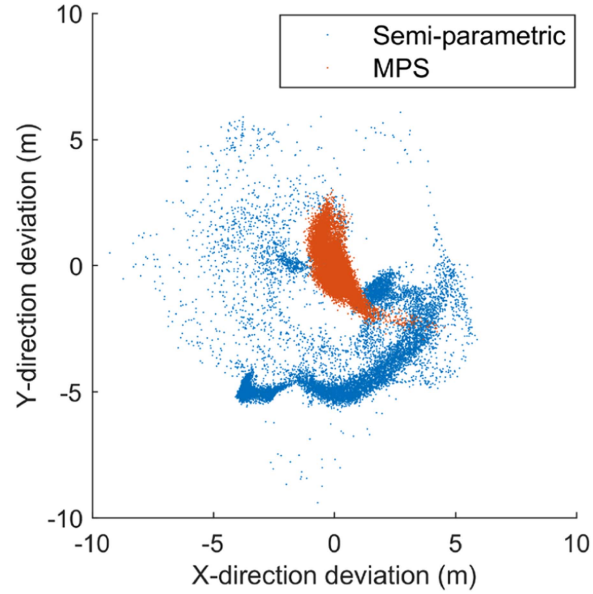

Fig. 10. Deviation value between static positioning result under the natural environment obtained through MPS and semiparametric method and standard position.

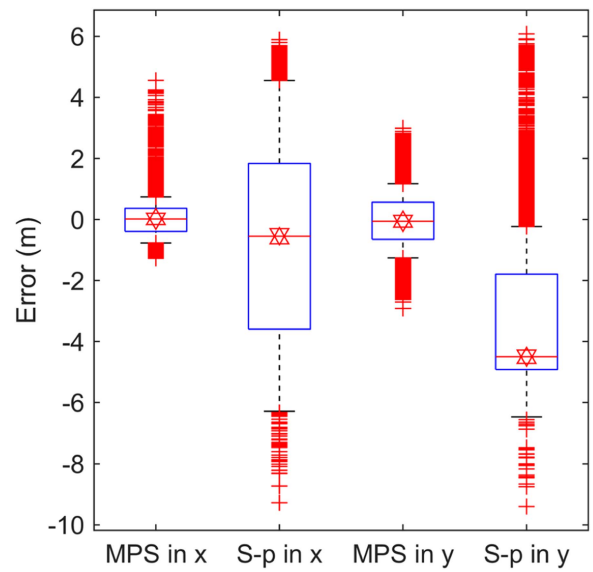

Fig. 11. Box diagram of static positioning error results under the natural environment of MPS and semiparametric method in X and Y axis.

running time of the semiparametric model is 0.53 ms, and that of MPS is 0.34 ms. This also shows that the MPS method has good computational efficiency, increasing the computational speed by about 35.85%. Notably, in terms of average efficiency, the MPS algorithm has a significant advantage, but this advantage is because it uses the physical characteristics of PDE, which reduces the solving frequency of PDE from each epoch to each communication state change. In other words, the higher computational efficiency is the advantage of the solution idea, not the benefit of the algorithm itself, and the algorithm still has significant time redundancy and case resource occupation due to its violent solution.

Research is underway to address this problem. It is proposed to solve the data of multiple epochs simultaneously under the time window and reconstruct the solution equation using the invariable PDE of various ages under the same communication state. Preliminary results have been obtained. Unfortunately, this method still needs to call the PDEC model on the first epoch. However, after removing the first epoch, the algorithm can reduce the attention of the SMA to the detection of the communication state and significantly reduce the interference of the PDEC model on the solution process. The solution efficiency has been dramatically improved.

E. Positioning Effect of the Indoor Positioning Algorithm Based on MPS Under the Natural Environment

To elucidate the generalizability of the MPS framework, an additional static experiment was conducted in a standard conference room measuring 21.5 m in length and 14.5 m in width. The investigation was instrumental in verifying the algorithm's performance across diverse scenarios. Fig. 9 presents the actual procedure, depicting the relative positions of the routers and the phone aligned with the ideal condition. Notably, the routers' positions in the figure are designated as AP1, AP2, AP3, and AP4.

1) *Static Experiment Results Under the Natural Environment:* In the static experiment, the location of the phone is point P in Fig. 9. The data upload period was set to 200 ms and the positioning results within 1 h were calculated under two methods (the LS algorithm based on MPS and the semiparametric model). The positioning results of the two methods are illustrated in Figs. 10 –12.

As depicted in Fig. 10, the proposed method demonstrates substantial improvement in positioning results compared to the semiparametric model in complex environments of the selected scenario. Both ways exhibited inferior performance when compared to the ideal environment. However, despite the challenges posed by the thorny environment, the positioning results of the proposed method still achieves convergence, albeit not reaching the same

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TABLE IV
Data Statistics of the Static Positioning Results Under the Natural Environment Were Obtained Using Two Methods

Precision / Method	ME		STD		RMSE	
	X	Y	X	Y	X	Y
Semi-parametric	-0.589m	-3.350m	2.815m	2.152m	2.876m	3.981m
SMA & PDEC	0.041m	0.029m	0.577m	0.901m	0.578m	0.901m
Promotion	93.04%	99.13%	79.50%	58.13%	79.90%	77.37%

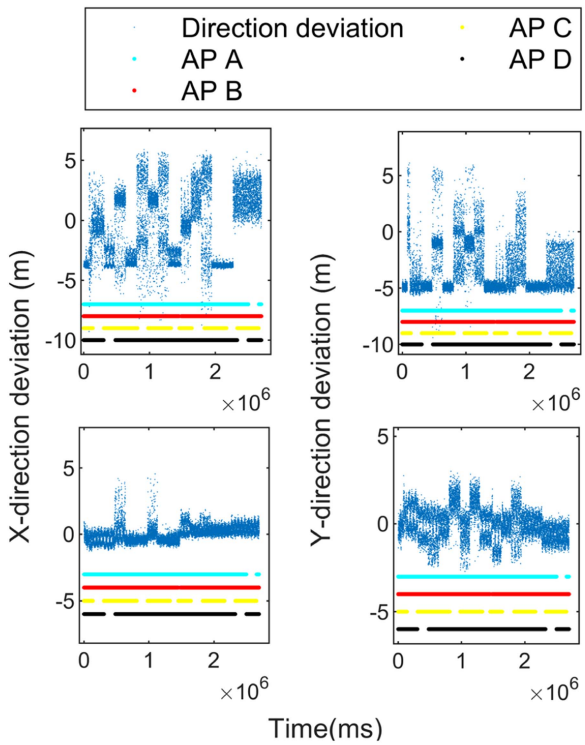

Fig. 12. Changes in the deviation of the static positioning results under the natural environment of the two methods over time. The left panel shows the deviation in the x-axis, while the right panel shows the divergence in the y-axis. And the upper panel shows the result of the semiparametric model, while the lower panel shows the effect of MPS.

level of accuracy as in the ideal scenario. This highlights the robustness and viability of the proposed method even under challenging conditions. Fig. 11 shows the positioning effects of the two methods. The localization stability of the proposed method is improved compared with the semiparametric model in this experiment. The maximum positioning errors of the MPS and semiparametric methods on the x-axis are 4.55 and 9.27 m, respectively; and the maximum errors on the y-axis are 2.99 and 9.40 m, respectively. Also, the semiparametric model has much more outliers than the proposed method. As Fig. 12 shows, the positioning effect of this method exceeds that of the semiparametric model in the x-axis and y-axis directions of each state period. Indeed, when the proposed method operates in complex environments, specific state periods may exhibit layering due to the influence of the environment. This phenomenon arises because, during static ranging, the signal propagation path remains unchanged, resulting in noticeable layering and trailing of the ranging results due to the multipath effect.

Consequently, the layering of ranging results also leads to corresponding layering and trailing effects in the positioning results. Such observations are essential to consider while evaluating the performance of the proposed method in challenging and dynamic environments where the multipath effect can significantly impact the positioning results. Table IV presents the ME, STD, and RMSE based on the semiparametric model and MPS algorithm. The proposed method showcases remarkable enhancements, exceeding 50% in all performance metrics when compared to the semiparametric model, underscoring the superior positioning accuracy and stability of the proposed method over the semiparametric approach.

2) *Dynamic Experiment Results Under the Natural Environment:* This section tested the positioning effect of the two methods (the LS algorithm based on MPS and the semiparametric model) under dynamic conditions. The experimental setting is consistent with the static positioning experiment in the conference room. A pedestrian held the mobile phone, and the data upload period was set to 500 ms. The pedestrian performed periodic and uniform linear motion in the conference room. The motion trajectory spanned from 0 to 8 m in the x-axis direction, corresponding to points A and B in Fig. 9. The MPS method and the semiparametric model were applied to estimate the location. The positioning results within 10 min were calculated under both methods.

Fig. 13 presents the positioning results under dynamic conditions for both methods. The MPS method accurately reflects the motion trajectory, aligning well with the real motion trajectory of the pedestrian, and it can be easily seen that the pedestrian's motion follows a straight line trajectory. Moreover, the dispersion of the positioning results in the y-axis direction is superior for MPS compared to the semiparametric model, indicating better positioning accuracy.

Fig. 14 depicts the temporal variations of the positioning results in the x and y-axes directions. For both MPS and semiparametric model, the positioning results exhibit periodic trajectories in the x-axis direction, reflecting the periodic motion of the pedestrian along the x-axis in a linear reentry path. The curves indicate that the trajectory change period corresponds to the period of a pedestrian's reentry motion. The maximum and minimum points are identified within each period as the two turning points in the pedestrian's reentry motion. The positioning results of

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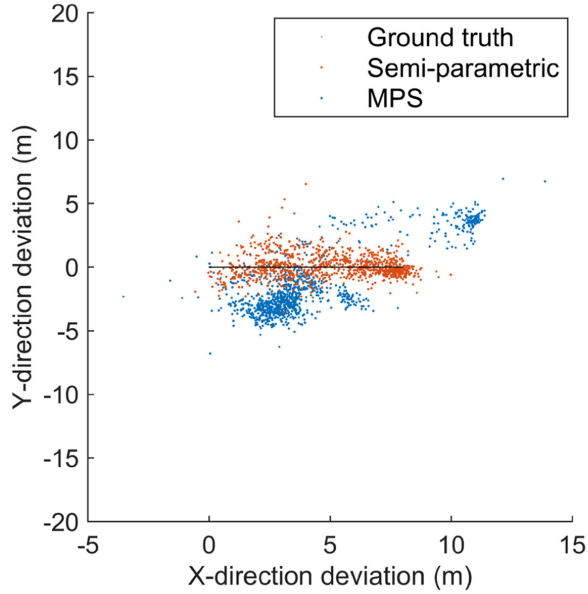

Fig. 13. Deviation value between dynamic positioning results under the natural environment obtained through MPS and semiparametric method and ground truth.

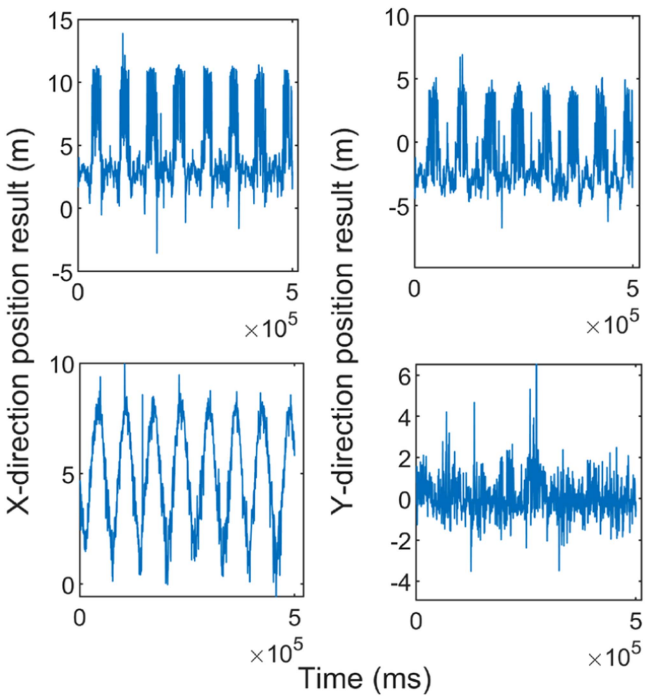

Fig. 14. Changes in the deviation of the dynamic positioning results under the natural environment of the two methods over time. The left panel shows the variation in the x-axis, while the right panel shows the divergence in the y-axis. And the upper panel shows the result of the semiparametric model, while the lower panel shows the effect of MPS.

MPS appear smoother and more straightforward, closely resembling the actual motion trajectory. However, the semiparametric model does not correct PDEs well, particularly evident at turning points where significant offset is observed. The positioning result in the y-axis direction reflects the mistake results in the order orthogonal to the pedestrian's moving direction, and the error results should be precisely

TABLE V
Data Statistics of the Dynamic Positioning Results Under the Natural Environment Were Obtained Using Two Methods

Precision / Method	ME	STD	RMSE
Semi-parametric	1.328 m	2.646 m	2.959 m
MPS	0.065 m	0.945 m	0.946 m
Promotion	95.11%	64.29%	68.03%

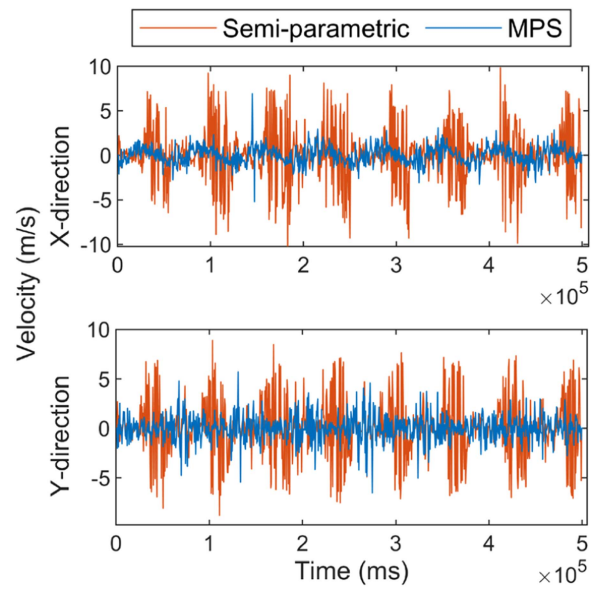

Fig. 15. Changes in the deviation of the dynamic velocity results under the natural environment of the two methods over time. The left panel shows the variation in the x-axis, while the right panel shows the divergence in the y-axis. And the upper panel shows the result of the semiparametric model, while the lower panel shows the effect of MPS.

zero in error-free conditions. In Fig. 14, the error of the semiparametric model in the y-axis direction also shows a certain periodicity. This is because the distortion error is uncorrected, and the positioning error will show periodic changes when the phone moves periodically. While the positioning error fluctuation of MPS results in the y-axis direction shows a random state, exhibiting lower levels of error compared to the semiparametric model, which indicates that the positioning accuracy and stability of MPS outperform the semiparametric model under dynamic conditions.

Table V shows the ME, STD, and RMSE based on the semiparametric model and MPS algorithm under the dynamic experiment. The values are calculated based on the error in the direction orthogonal to the motion trajectory. In terms of both accuracy and stability, MPS is superior to the semiparametric model, and the improvement effect of each index is more than 60%.

To further corroborate the effectiveness of MPS, Figs. 15 and 16 depict the variations in velocity and acceleration of the mobile phone overtime during the dynamic experiment. The experimental outcomes also confirm that the pedestrian

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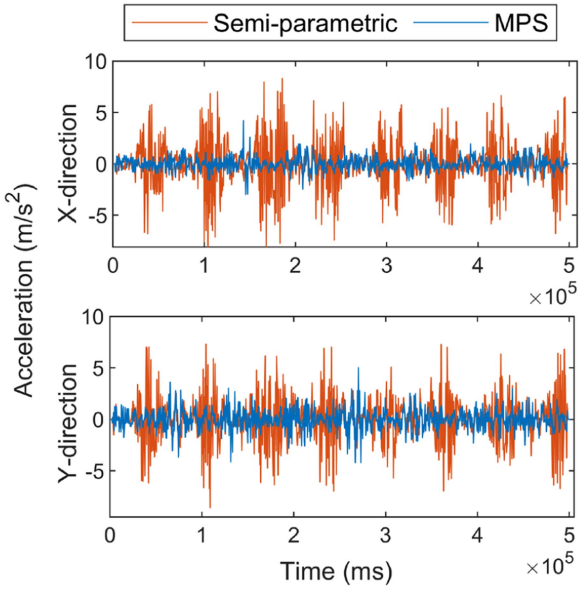

Fig. 16. Changes in the dynamic acceleration deviation results under the two methods' natural environment over time. The left panel shows the deviation in the *x*-axis, while the right panel shows the divergence in the *y*-axis. And the upper panel shows the result of the semiparametric model, while the lower panel shows the effect of MPS.

is in periodic and uniform linear reentry motion. During the investigation, although the pedestrian endeavors to maintain constant motion, there are still pronounced occasional fluctuations in velocity due to movement, which reflects the authenticity of the results data and the correctness of the method.

In brief, although this method has some limitations, it can effectively improve the accuracy, calculation speed, and stability of WIFI-RTT positioning. This method has strong practicability and a significant advantage in real-time application.

IV. CONCLUSION

First, this article considers the WIFI-RTT ranging principle and analyzes the influence of the data upload period on ranging results. On this basis, an MPS framework is proposed, a reference template for the positioning system that can be obtained based on the CD model. Then, an SMA and a PDEC model suitable for an MPS are proposed to capture the moment of communication state change by the SMA, i.e., the moment of PDE reset is quickly captured, and then, the PDE is corrected by the PDEC model at the initial stage of communication state change. The experimental results demonstrate that the SMA can provide a success alarm rate of more than 96%, and the success alarm rate increases as does the data upload period. However, the SMA has a poor alarm effect on the communication state change caused by the cumulative time difference exceeding the FTT. The PDEC model can deal with PDEs better than the semiparametric model, but its disadvantage is its efficiency. Therefore, how to reduce the efficiency of the PDEC model is the focus of future research. In addition, this framework contributes very little to the correction of the multipath effect, router layout, and ranging error caused by noise. Research on the multipath effect, router layout, and noise error will be conducted in the future.

DATA AVAILABILITY

The data used to support the findings of this study are available from the corresponding author upon request.

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